

1.A

```
1  ADD  R4, R1, R2
2  LW   R3, 0(R9)
3  SUB  R3, R3, R5
4
5  LOOP:
6  ORI  R7, R3, 4
7  ADDI R5, R3, 1
8  AND  R8, R3, R7
9  LW   R6, 0(R7)
10 SW   R6, 0(R6)
11 OR    R4, R6, R3
12 ADDI R5, R6, 1
13 AND  R1, R4, R6
14 LW   R9, 0(R5)
15 SUBI R9, R9, 4
16 SW   R9, 0(R1)
17 SLT  R18, R15, R9
18 BEQ  R18, R9, LOOP
```

Listing 1: Original code

After adding NOPS:

```
1  ADD  R4, R1, R2
2  LW   R3, 0(R9)
3  NOP
4  NOP
5  SUB  R3, R3, R5
6  LOOP:
7  ORI  R7, R3, 4
8  ADDI R5, R3, 1
9  NOP
10 AND  R8, R3, R7
11 LW   R6, 0(R7)
12 NOP
13 NOP
14 SW   R6, 0(R6)
15 OR    R4, R6, R3
16 ADDI R5, R6, 1
17 NOP
18 AND  R1, R4, R6
19 LW   R9, 0(R5)
20 NOP
21 NOP
22 SUBI R9, R9, 4
23 NOP
24 NOP
25 SW   R9, 0(R1)
26 SLT  R18, R15, R9
```

```
27 NOP
28 NOP
29 BEQ  R18, R9, LOOP
```

Listing 2: Code with NOPS

Total: 16 NOPS

$$t_c = 2_{ns}, ICU = 1, t_i = ICU \cdot t_c = 2_{ns}$$

$$IC = 30, CPU_{time} = IC \cdot t_i = 30 \cdot 2_{ns} = 60_{ns}$$